GOOGLE DATA ANALYTICS CAPSTONE PROJECT

Cyclistic Bike Share Analysis

Prepared by **Abhirami Ananthakumar**

TOOLS

Google Sheets • R • Tableau • Google Docs • Google Slides • GitHub

2026

TABLE OF CONTENTS

Agenda

SECTION 01

Business Problem

Annual Members

Casual Riders

- Cyclistic is a bike-share company serving riders through two membership options: annual members and casual riders.
- Casual riders use the service frequently but are less likely to purchase annual memberships.
- Increasing annual memberships is important because they provide more predictable, long-term revenue.

Business Objective

Analyze 12 months of Cyclistic ride data to identify behavioral differences between annual members and casual riders.

Provide data-driven recommendations to increase annual memberships.

SECTION 03

Business Task

Provide actionable insights that help Cyclistic's marketing team convert more casual riders into annual members.

SECTION 04

Dataset

SOURCE

Divvy Trip Data (Cyclistic)

PERIOD

Jul 2025 – Jun 2026 (12 months)

TOTAL RECORDS

5.93 Million Rides

FORMAT

Monthly CSV Files

Data cleaned and combined using R

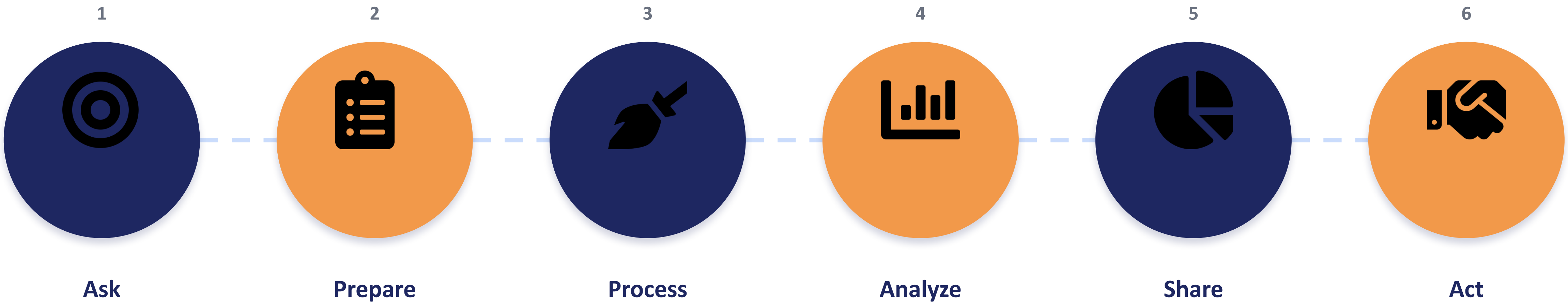
Tools Used

TOOLS	PURPOSE
Google Sheets	Initial data exploration
R	Data cleaning & preprocessing
Tableau	Dashboard creation
Google Docs	Project Documentation
Google Slides	Presentation
GitHub	Portfolio & version control

SECTION 06

Methodology

The Google Data Analytics six-phase framework



SECTION 07

Data Cleaning

- Imported 12 monthly datasets
- Combined datasets into one file
- Removed duplicates
- Removed missing values
- Converted date and time columns

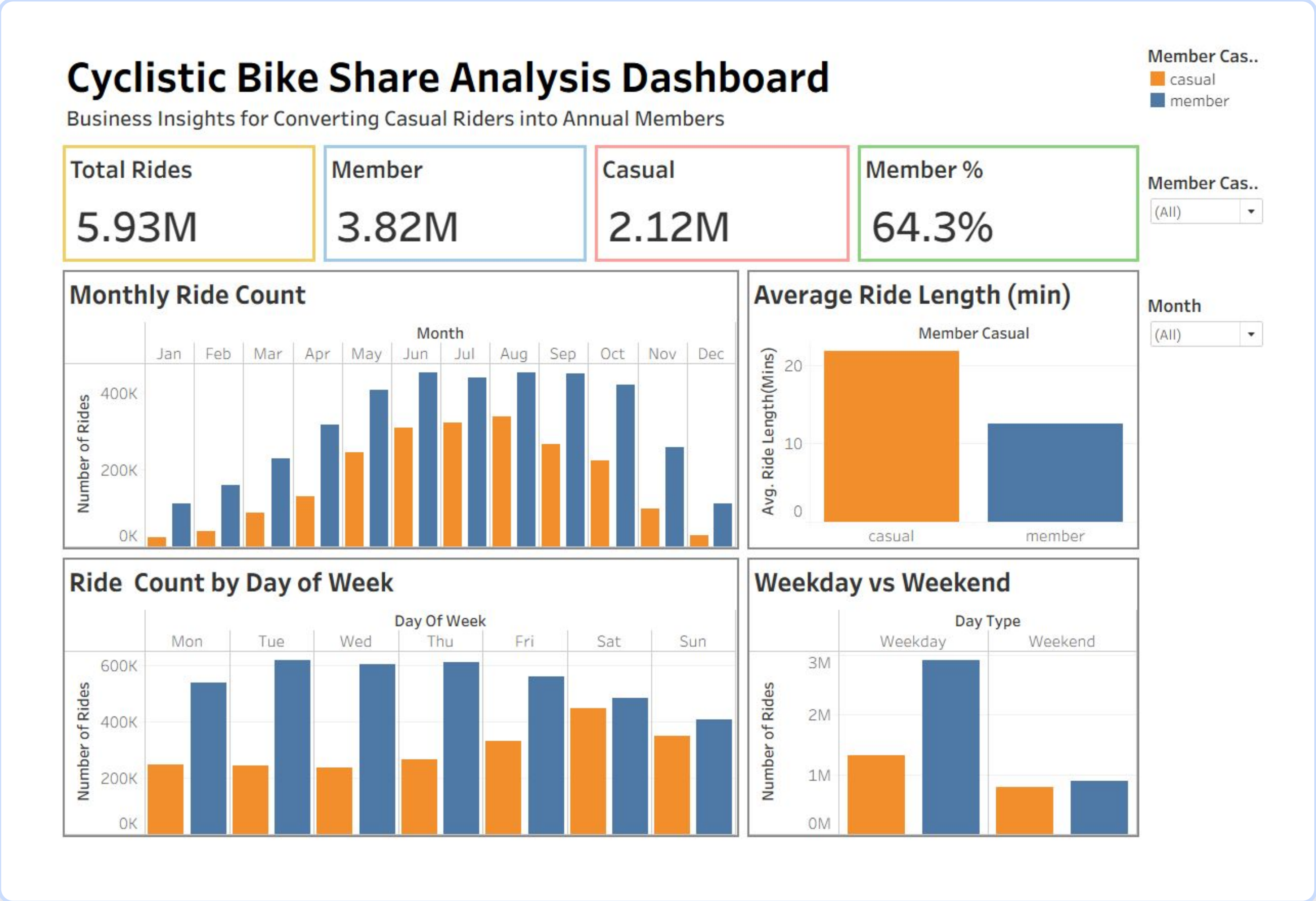
NEW VARIABLES CREATED

- Ride Length
- Day of Week
- Month
- Hour

SECTION 08

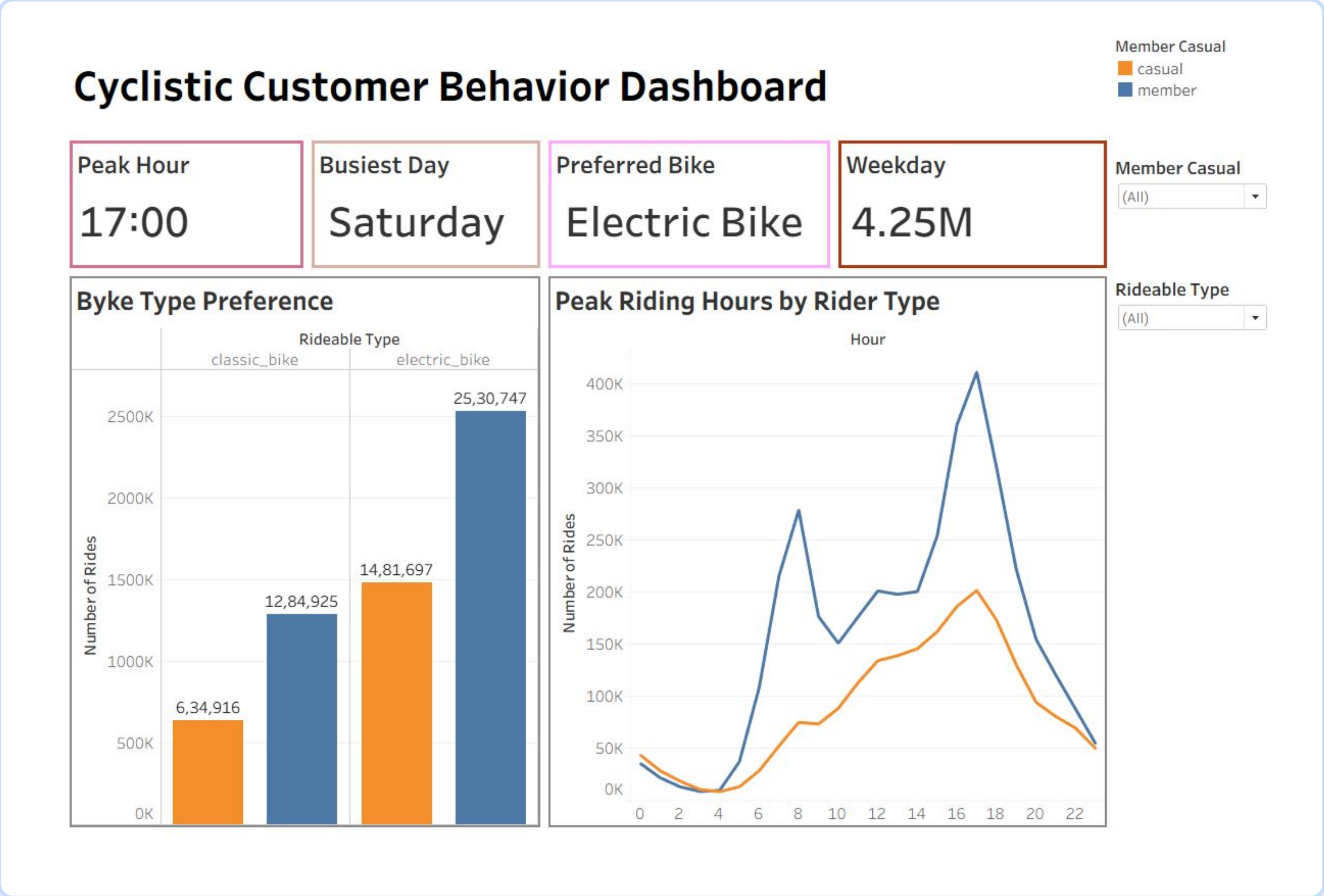
Dashboard Overview

Executive Summary Dashboard



Dashboard Overview

Customer Behavior Dashboard



SECTION 09

Key Findings

- Casual riders take longer rides.
- Members ride more frequently.
- Casual rides increase during weekends.
- Members ride consistently throughout the week.
- Summer records the highest number of rides.
- Peak riding hour is 5:00 PM.
- Electric bikes were popular among both annual members and casual riders.

Business Recommendations

01

Weekend membership promotions

02

Electric bike membership benefits

03

Summer marketing campaigns

04

Evening benefits promotions

05

Commuter membership plans

SECTION 11

Expected Business Impact

Increase annual memberships

Improve customer retention

Enable data-driven marketing decisions

Skills Demonstrated

- Data Cleaning
- Data Transformation
- Data Analysis
- Data Visualization
- Business Intelligence
- Data Storytelling
- Tableau Dashboard Development
- R Programming

SECTION 13

Conclusion

This project analyzed 5.93 million Cyclistic rides to understand behavioral differences between annual members and casual riders. The insights provide practical recommendations that can help increase annual memberships and support business growth.

THANK YOU

Cyclistic Bike Share Analysis • Abhirami Ananthakumar